

// AUTONOMOUS SPACE OPERATIONS

DRIFT 3D

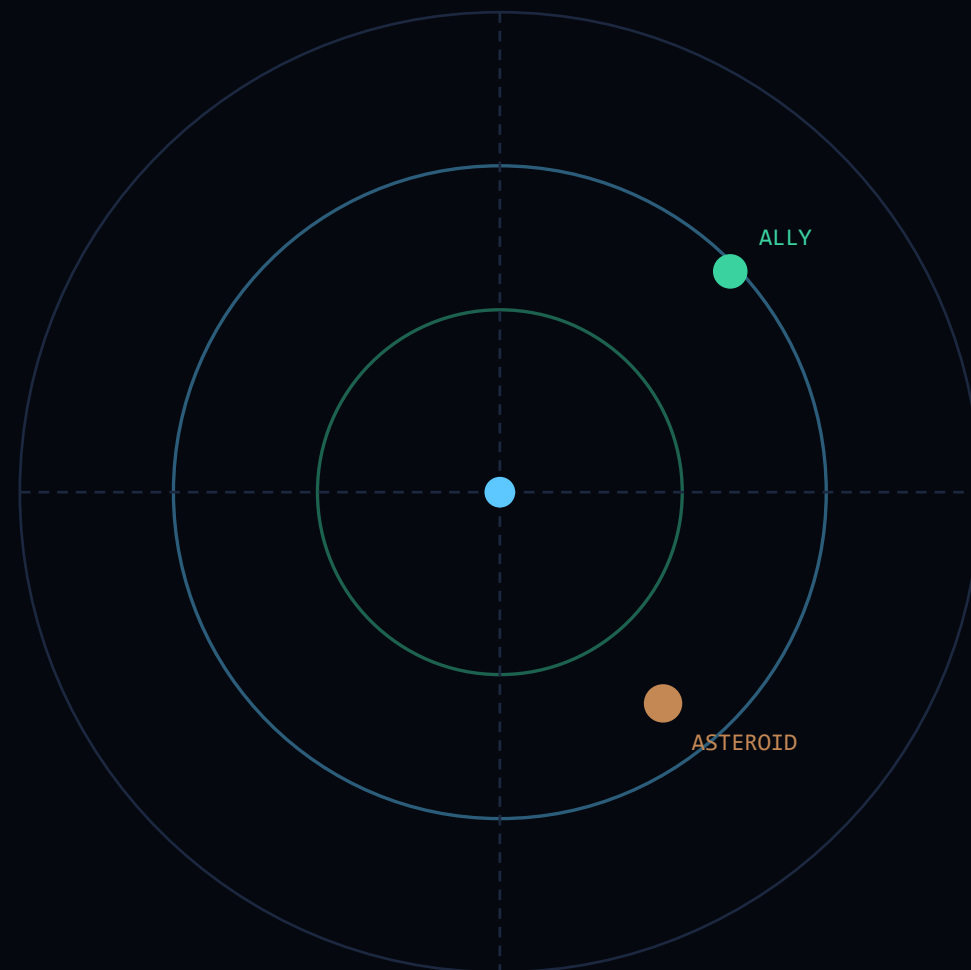
Autonomous Conjunction Resolution on Electric Propulsion

*An agent that perceives, plans, and flies real physics —
inside a world dreamed in real time.*

PROTECT

AVOID

LEAST Δv



Speaker: Space is getting crowded. Over 30,000 tracked objects, and satellites now dodge each other tens of thousands of times a year. DRIFT 3D is an interactive simulation of an autonomous probe doing exactly that — protecting an ally, avoiding a hazard, on real propulsion limits.

A converging encounter, two constraints



Protect the ally

A non-maneuvering satellite must stay outside the safe ring.



Avoid the asteroid

An incoming hazard on a collision course with the probe.

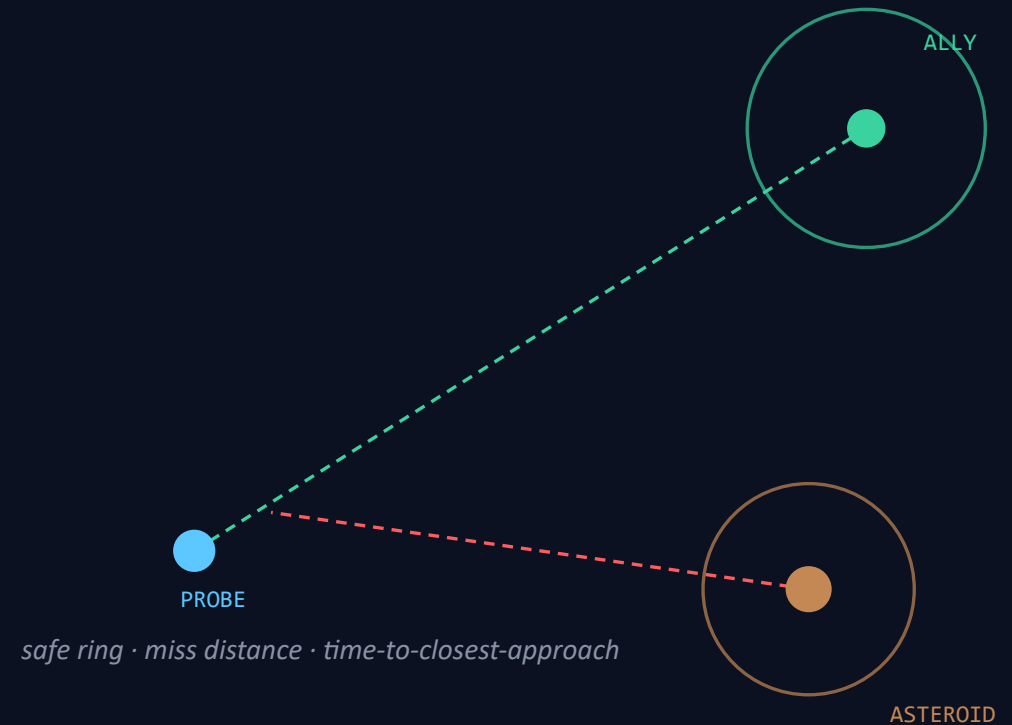


Spend the least Δv

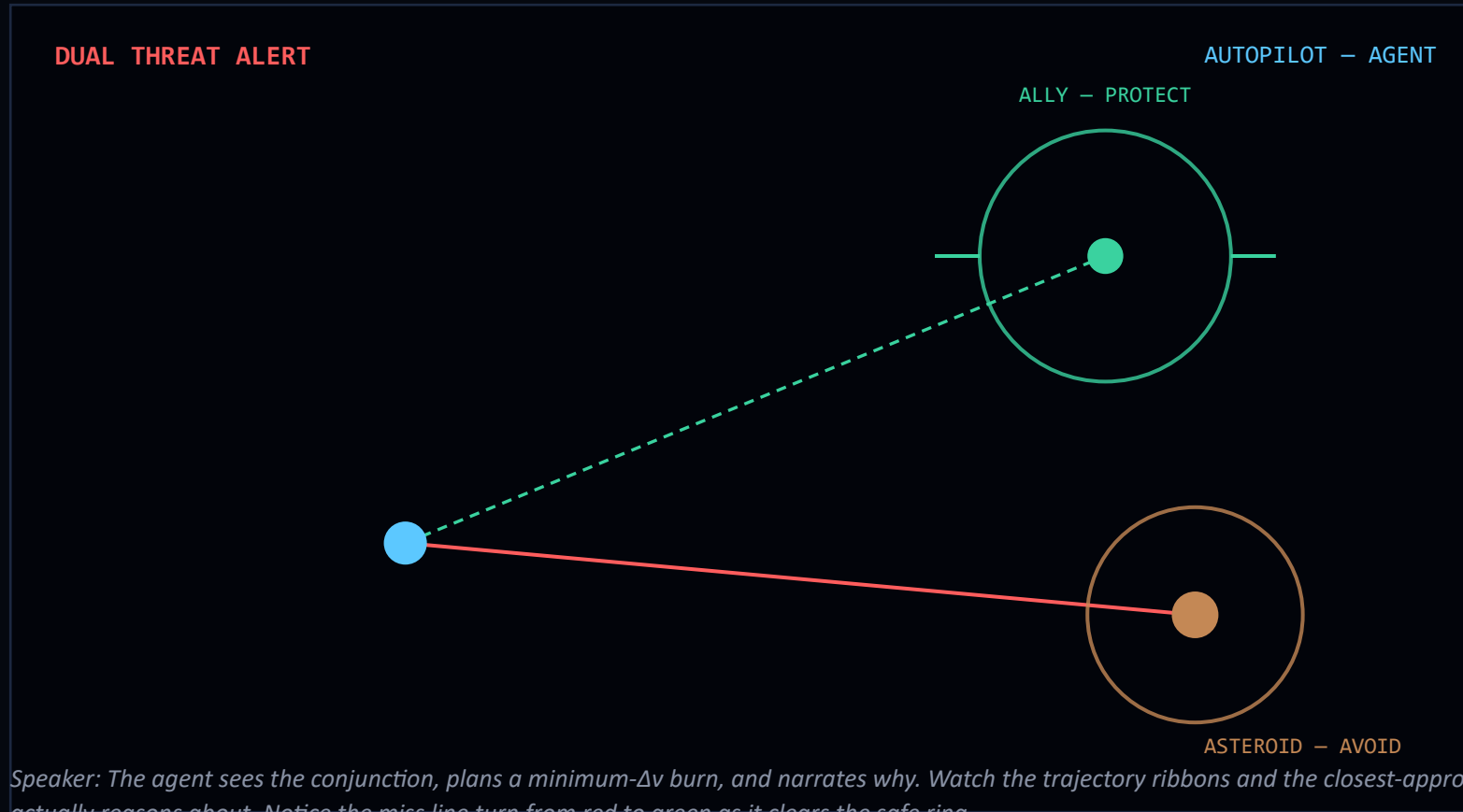
Every maneuver burns propellant — and propellant is mission life.

Speaker: This is the real operational dilemma: you get a close-approach warning, and moving costs fuel you can never get back. The skill isn't dodging — it's dodging cheaply, without endangering something else.

CONJUNCTION GEOMETRY



Watch the agent resolve a conjunction



WHAT TO WATCH



Trajectory ribbons trace each body's predicted path.



TCA markers pin the moment of closest approach.



The miss line flips red → green as it clears the safe ring.



The Δv ledger grades efficiency against the optimal burn.

Electric propulsion is the law

Tiny thrust, huge efficiency — heavy & committed: plan early, coast for free.

178nN

thrust

3,200s

specific impulse

31.4m/s

exhaust velocity

×630

time-warp to play

Real ion engines are agonizingly gentle — millinewtons of thrust. The craft is heavy and committed: plan turns early, ration xenon, and **coast for free**. Real ion burns take minutes; the time-warp compresses reality so it's playable.

Speaker: Real ion engines are agonizingly gentle — millinewtons. The console shows honest numbers; the 'time-warp' is us compressing reality so it's playable. The lesson stays true: you cannot stop or turn on a dime.

The world changes what the craft can do



IN SUNLIGHT

100% power authority

THRUST AUTHORITY



Full thrust available — the agent can maneuver freely.



IN SHADOW

≈30% power authority

THRUST AUTHORITY



Thrust fades — the agent eases off and coasts until it brightens.

Thrust power is read from the brightness of the generated scene — exactly like a solar-electric satellite losing power in eclipse.

Speaker: Thrust power is read from the brightness of the scene. Fly into shadow — like a real satellite entering eclipse — and solar-electric power drops, thrust weakens. The agent eases off and coasts until it brightens.

Grab the stick, hand it back



AGENT

Autopilot

Plans and flies the minimum- Δv solution by default.



YOU

Manual (press G)

Take over to dodge by hand, then release control back.

Δv LEDGER



■ agent ■ you
live agent country is graded A–F on Δv efficiency vs. the optimal burn.

Autonomy isn't all-or-nothing — the human-on-the-loop model real operations are moving toward.

Speaker: Autonomy isn't all-or-nothing. Press G and you fly manually; release and the agent resumes. The Δv ledger tracks who spent what — agent vs. you — and grades the result against the optimal.

Maps to today's space operations

30,000+

tracked objects in orbit

30+

ISS debris-avoidance maneuvers

tens of thousands

Starlink maneuvers / 6 months



Conjunction screening

The US Space Force issues Conjunction Data Messages keyed on TCA & miss distance — the same quantities in the game.



Automated avoidance

ESA and others are building AI-assisted collision-avoidance as object counts climb toward Kessler-syndrome risk.

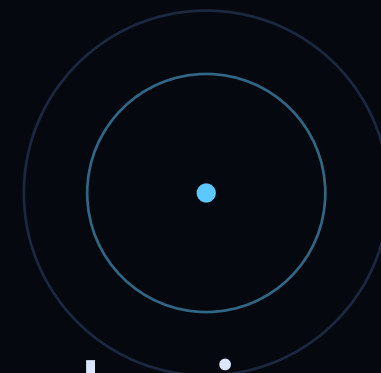


Solar electric propulsion

Ion & Hall thrusters fly today on Dawn, Starlink and GEO comsats — solar-powered, so they lose thrust in eclipse.

Speaker: Every concept on screen is a real one. As object counts climb toward Kessler-syndrome risk, autonomous avoidance under propellant constraints stops being a game and becomes infrastructure.

// CLOSE



An agent that perceives, plans, and flies real physics — inside a world dreamed in real time.

Deterministic physics you can trust · A generative world you can't predict · An autonomous pilot reasoning between them.

DRIFT 3D

Thank you — happy to take questions.

Speaker: Deterministic physics you can trust, a generative world you can't predict, and an autonomous pilot reasoning between them. That's DRIFT 3D.